18/01/26
26/01/26 [edited]

Problem statement: Conventional PID Controllers show reduced Performance of PMSM speed Control under varying load parameter Changes. This project aim to design a fuzzy logic Controller to improve Speed regulation and dynamic response.

Problem statement questions.

1. what is PMSM motor ? its working principle ? speed control methods
2. what is the drawback [or] what we want to improve?
3. why PMSM or PID sensitive to load parameters?
4. what are PID controllers?
5. How we gonna improve the drawback?
6. what is fuzzy ? and we are using fuzzy in this project?
7. How fuzzy method is efficient than other control methods?
8. How this method solved the drawback?

22/01/26 - 27/01/26

Overview of project

1. Concept: Complete theoritical clarity over the the project
2. Designing : Designing Both
   i. PID controller
   ii. Fuzzy controller

   PID controller : To prove the problem what we are trying to do exist
   Fuzzy controller : To solve the problem
3. Simulation : Executing Both controller Circuits in Matlab Simulink
4. Hands in loop : Once Completion of Matlab simulation starting the work in d space or OPAL-RT

Phase 01: Before mid [9/3/26]

1. Concept
2. Designing
3. Executing in matlab and Completing Simulation

Phase 02: Before week of last teaching day [04/05/26]

4. Executing in dspace or OPAL-RT
5. changes and Completing hands in loop
6. Report submission

details

1. Concept: 1.1] Theory of PMSM
   1.2] Speed controllers
2. Design 2.1] design PID controller
   For testing i] under normal conditions
   ii] load torque variations
   iii] parameters changes
   To show that i] slower dynamic response
   ii] Overshoot
   iii] Steady state response
   2.2] Now designing Fuzzy controller
   i] Faster settling
   ii] Reduced overshoot
   iii] steady state response
3. Matlab Simulation 3.1] Circuits performed in matlab
   3.2] Comparitive Simulation analysis
   To show i] Rise time
   ii] settling time
   iii] overshoot
   iv] speed ripple under load change
4. Hands in loop: 4.1] Prepare the models to execute in dspace
   4.2] Changes in mode
   4.3] Real time execution with input and output.
   4.4] test i] load changes
   ii] observing speed response
5. Report submission: Real time implementation will be demonstrated using simulation on OPAL-RT / d space.

Report:

1. Problem statement:
2. Equipments and softwares Used:
3. Objective about the experiment
4. designing
5. Matlab simulation and procedure
6. Comparison observations
7. Hands in loop
8. Final observations
9. Proof of working
10. Results

Week 01 Research paper Under Standing
1. Select a paper
2. Study the paper
3. Understand the concept

Week 02: Problem statement
1. Come up with problem statem
2. what is drawback
3. How we are solving

Week 03: PID controller design
1. Design PID controller
2. Standard model.

Week 04: PI Simulation Analysic
1. Equation derivation , value obtain from graph
2. Values

Week 05: Fuzzy controller Design
1. Design fuzzy controller
2. Standard matlab designing

Week 06, 07, 08: Fuzzy controller Simulation
1. Equation derivation , value obtain from graph
2. values

Week 09: Performance Comparison table
1. PID vs Fuzzy

Week 10: Conference Paper Preparation
Converting project into a Conference Paper

01. PMSM mechanical equation without load

$$J\frac{d\omega}{dt} + B\omega = T_e - T_L$$

where J = inertia
B = viscous friction
ω = angular speed
T_e = electromagnetic Torque
T_L = load torque
Motor $T_e > T_L$

[$T_L = 0$ at no load]

$$\Rightarrow J\frac{d\omega}{dt} + B\omega = T_e$$

[Take laplace Transform]

$$\Rightarrow Js\,\omega(s) + B\omega(s) = T_e(s)$$

$$\Rightarrow \omega(s)[Js+B] = T_e(s)$$

$$\Rightarrow \frac{\omega(s)}{T_e(s)} = \frac{1}{Js+B}$$

$\left(\frac{C(s)}{R(s)}\right.$ I/p → Torque, O/p → Speed$\left.\right)$

02. Type of System [1st or 2nd]

$$\frac{\omega(s)}{T_e(s)} = \frac{1}{Js+B} = \frac{\frac{1}{B}}{\frac{J}{B}s+1} = \frac{K}{\tau s+1}$$

Standard 1st order

$\frac{K}{\tau s+1}$ (gain) $K = \frac{1}{B}$, $\tau = \frac{J}{B}$

τ — Time constant

∴ J↑ τ↑ response speed↓

Note: In no load condition, the pmsm mechanical eq.n gets reduced to 1st order system so J (inertia) plays a major performance in time response.

03. PMSM Mechanical equation to TF.

$$J\frac{d\omega}{dt} + B\omega = T_e - T_L$$

$$\Rightarrow Js\,\omega(s) + B\omega(s) = T_e(s) - T_L(s)$$

$$\Rightarrow \omega(s)[Js+B] = T_e(s) - T_L(s)$$

$$\omega(s) = \frac{T_e(s)}{\omega(s)} - \frac{T_L(s)}{\omega(s)}$$

∴ Now R(s) are $T_e(s)$ & $T_L(s)$ Both

∴ Note: with load included, the model becomes two input single output system. The load torque is like negative disturbance not a controller feedback.
- acts opposite to motion
- it subtracts from speed
- But its not a negative feedback, PID reacts to feedback so now PID doesn't react to this directly but only reacts when speed drops (which is because of the T_L)

Analogy T_e = accelerator of car
T_L = wind resistor not brake feedback.

T_e	T_L
1. control I/p	1. disturbance I/p
2. Generated by inverter & current control	2. Its from mechanical load etc.
3. we can control	3. we can't control

Final TF

i) T_e to Speed $\frac{\omega(s)}{T_e(s)} = \frac{1}{Js+B}$
control input

ii) T_L to speed $\frac{\omega(s)}{T_L(s)} = -\frac{1}{Js+B}$
disturbance input

Note: → 1st order
→ $\tau = \frac{J}{B}$
→ $K = \frac{1}{B}$

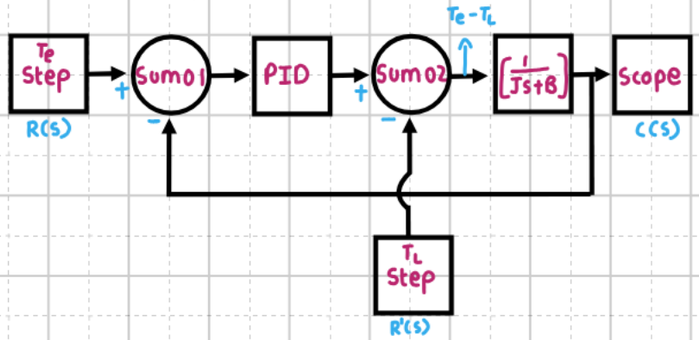
observations
1. T_L ↑ S↓.
2. T_e must ↑ to compensate.
3. Through speed control loop.

Note: With load torque included, the PMSM Transfer function becomes 1st orde with disturbance

04. Time response parameters

1. Rise time (T_r)
2. Settling Time (T_s)
3. Overshoot (M_p)
4. Steady state error (E_{ss})
5. Disturbance Rejection (DR)

05. Simulation values



J = 0.01; ↓J response↑
B = 0.1; ↓B ↓damping

05. values from Simulation

- T_e → constant
- T_L → vary
- P = 100 I = 50 D = 0.001

Step Time = 0 step time 0.2

S.No	T_e	T_L
01.	100	0
02.	100	20
03.	100	30
04.	100	40
05.	100	50

Simulation graph

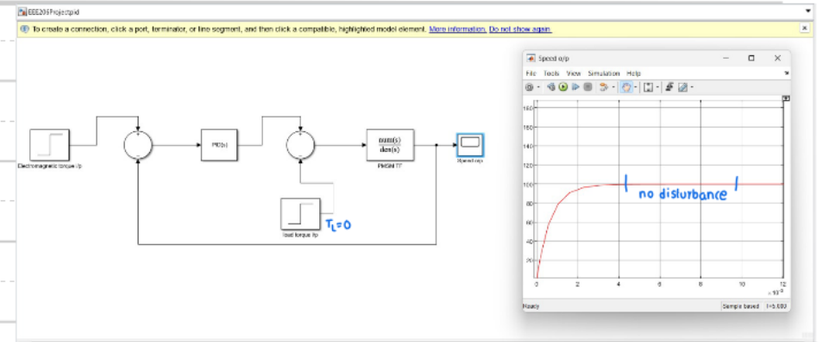
PID Equation: $u(t) = K_p e(t) + K_i \int_0^t e(\tau)d\tau + K_d \frac{de(t)}{dt}$

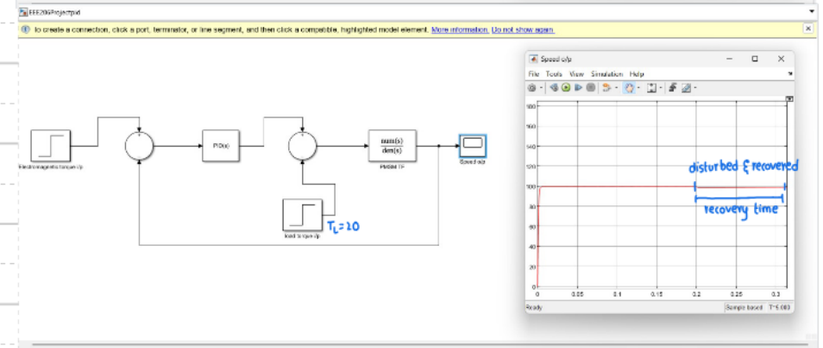
Step settings
Step time : 0
Initial V : 0
Final V : 100
Sample T: 0
This is where we are givin out desired speed
like at t=0 the speed is 0rpm now at t we want t=100rpm

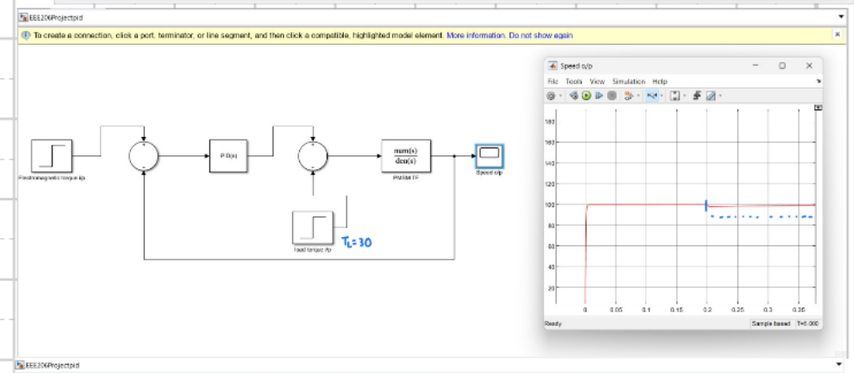
Sum:
The sum is mainly used to measure the error, like we we will keep our desired value but we dont know whether we are getting it or not so that's why we take feedback

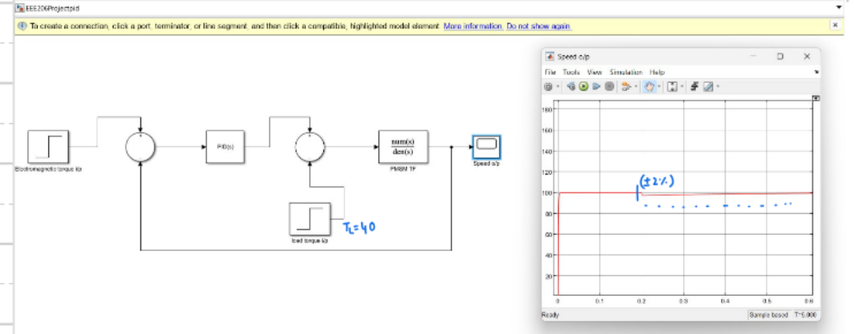
Error calculation
we will give input and we will get a o/p so the o/p is correct or not we will take feed back

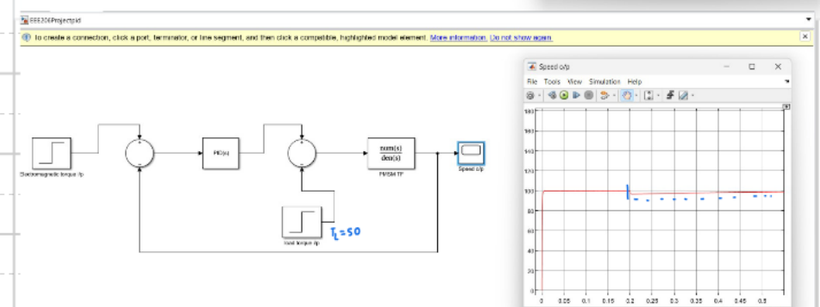
error should be
+ desired O/p − actual O/p = 0
+ value (+ve error)
− value (−ve error)



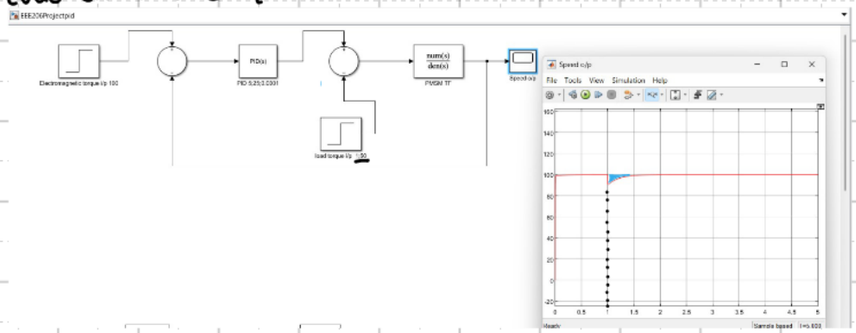




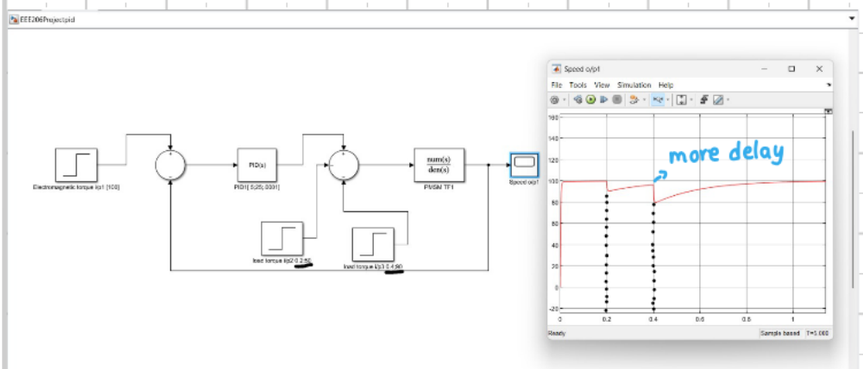




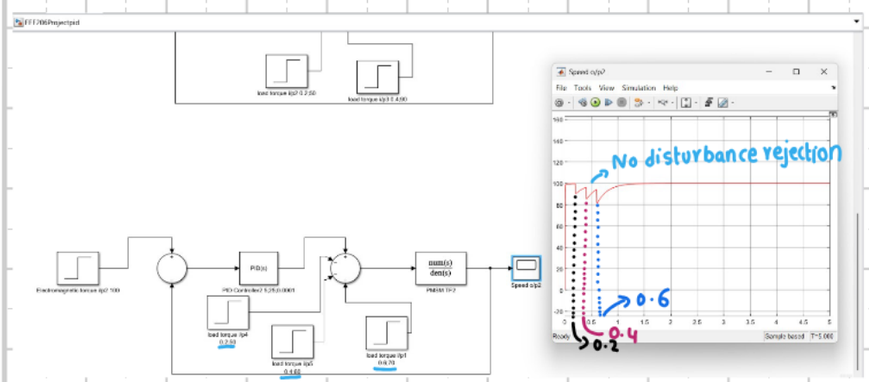
Load disturbance $T_L = 50$ at time: 1



varying load time to time



Fast & several load varying



Calculations for 3rd Model

values

1. $T_e = 100$ rad/s ; $T_L = 50$ rad/s at steptime = 0.2 s
2. $T_e = 100$ rad/s ; $T_L = 60$ r/s at s.T = 0.4 s
3. $T_e = 100$ r/s ; $T_L = 70$ r/s at s.T = 0.6 s

T_r, T_s, T_d, E_{ss} for T_e

Speed dip, (disturbance rejection / Recovery time)

ω_n = natural frequency ζ = damping ratio

1] Reference tracking performance.

Rise time [T_r]

10% to 90% final value

T_r =

$T_r = 0.005$ s

Delay time :

50% of final value

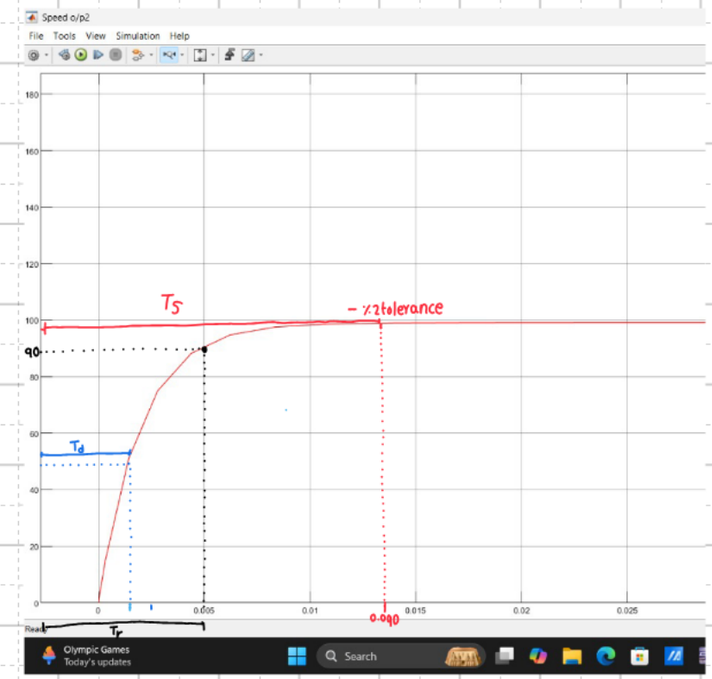
$T_d = 0.00152$ s

Settling time:

Tolerance ±2% [98 or 102]

-2% tolerance $T_s = 0.090$ s

$E_{ss} \approx 0$ for PID



2] Disturbance rejection performance

speed dip

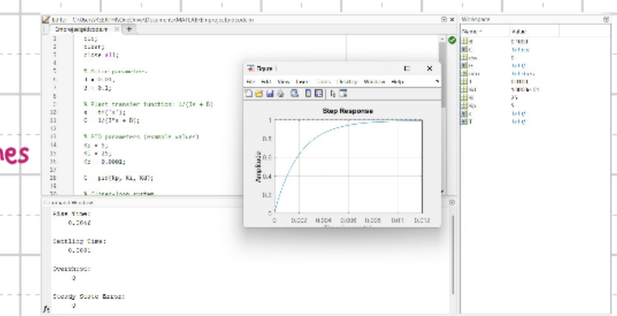
Minimum speed dip after disturbance

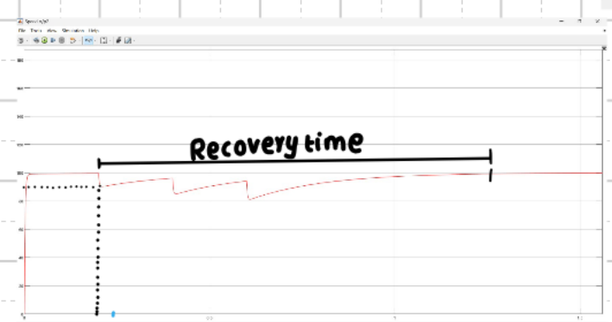
speed dip = 19 rad/s

Recovery time after apply 3 loads at 3 times

±2% tolerance

$T_R = 1.25 - 0.19 = 1.06$ s





25/02/26

1) What is fuzzy?
2) What is fuzzy vs PID?
3) Fuzzy Block specifications? & rule based
4) Simulation
5) Time response parameters.

Fuzzy Controller
i. Fuzzication
ii. Rule Base
iii. Inference Engine
iv. Defuzzification

Rule Base Table

e \ Δe → ↓	NB	NS	Z	PS	PB
NB	NB	NB	NM	NS	Z
NS	NB	NM	NS	Z	PS
Z	NM	NS	Z	PS	PM
PS	NS	Z	PS	PM	PB
PB	Z	PS	PM	PB	PB

output command

PID equation to fuzzy equation

$u(t) = PID = k_p e + k_i \int e\,dt + k_d \frac{de}{dt}$

$u(t) = \text{fuzzy} = f(e, \Delta e)$

What is fuzzy?

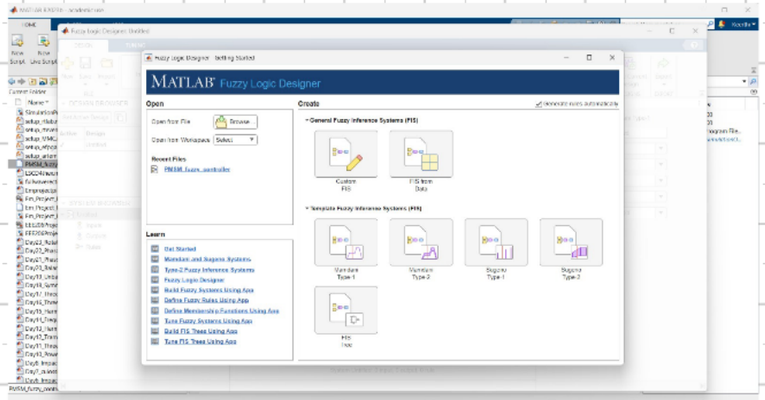
Explanation through a problem

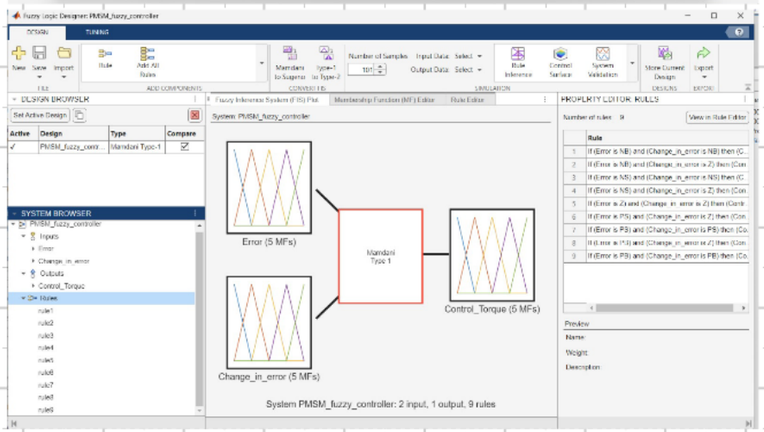
1. desired speed : 100 rpm
2. actual motor speed : 80 rpm
3. Error = Reference − actual 100 − 80 = 20 rpm (e)
4. Assuming previous error : 25 rpm
5. Δ Error $\Delta e = e(k) - e(k-1)$
   $= 20 - 25 = -5$ rpm
   e is reducing, so the motor is catching up the reference.
6. Fuzzification
   number = linguistic values
   Error = 20 → positive medium
   −5 → Negative small
   e = +ve medium
   Δe = −ve small → o/p = Z
   medium
   ↳ motor is slow
   ↳ but its already coming up
3. Defuzzification
   linguistic rule → Number
   medium so → o/p Torque = 3 Nm
   Nm → Torque, rpm → speed
   $\omega = \frac{2\pi N}{60}$ rad/s
8. Motor response
   Speed ⇒ 95 rpm → 100 rpm
   ∴ The e becomes small and fuzzy reduces torque automatically
9. Better than PID
   PID "uses a strict equation"
   fuzzy "creates rule based on the requirement"

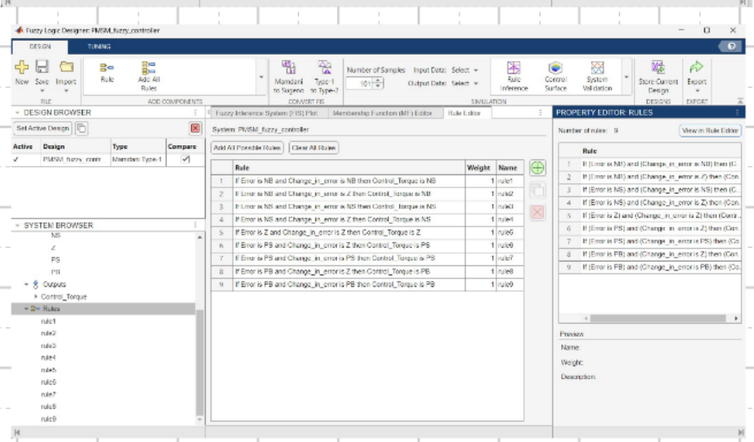
Ex:-
Fuzzy: Motor too slow → strong torque
almost correct → reduce torque

Rule based file edit
PMSM_fuzzy_controller.fis (fuzzy interface system file)
↳ either code or manual setup

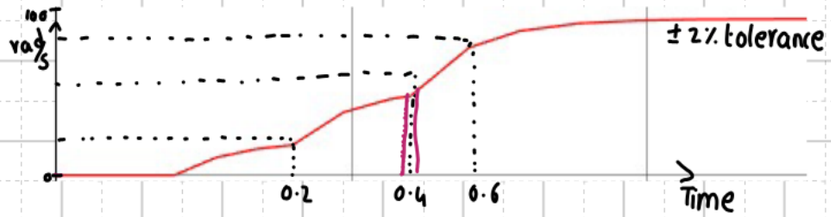
Manual setup
command window:
FuzzyLogicDesigner







Calculation

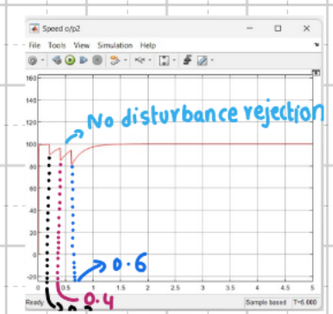
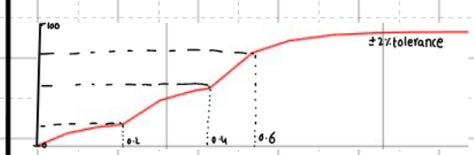


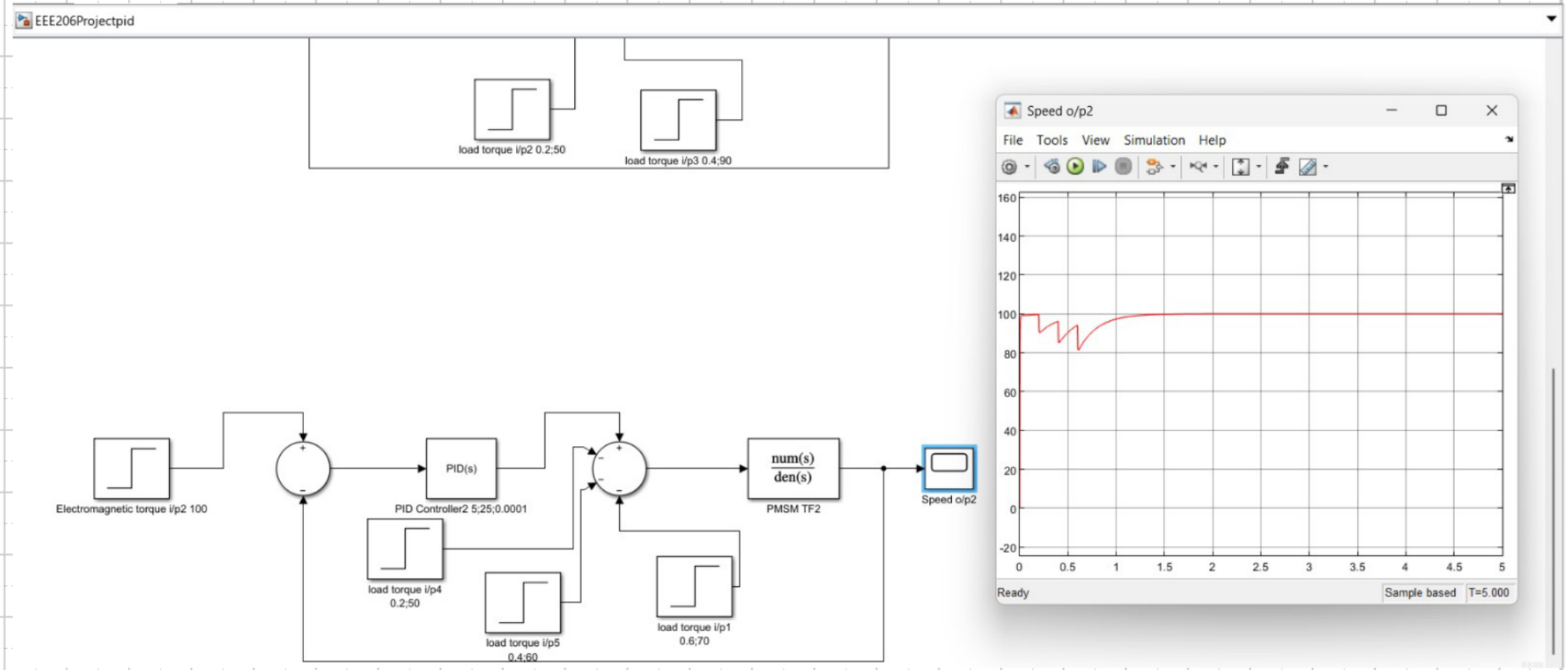
1. Speed dip: (max)
   = 50 - 48
   = 2 rad/s

2. Disturbance recovary
   0.043 - 0.041
   = 0.002 S
   = 2×10^{-3} S

Note: fuzzy have the adaptive tunning which makes the disturbance occurs before reaching steady state with ±2% tolerance

PID vs fuzzy comparison

PID	FUZZY
1. The core principle of proportional Integral derivative control is that they work on fixed mathematical model "fixed tuning" 2. Since PID values are fixed it struggles with unexpected load disturbances & system variations. leading to marked as a instable controller. 3. High initial overshoot 4. slower reaction to sudden load changes "disturbance rejection / recovery time" 5. Increased torque ripples under faulty conditions.	1. fuzzy works on rule based logic like "if-then" and also it doesn't required fixed model. "no fixed tuning. 2. fuzzy adapts the load disturbances and system variations since its not a fixed model. causing that as a flexible control system 3. Reduced overshoot, providing smooter start up Performance 4. Achieves faster settling time "disturbance rejection / recovery time" 5. Reduces ripples (G_m Torque) - adapative Tuning: 1. adjust P, I, D in real time according to current error and its change - Han
$u(t) = K_p e(t) + K_i \int e(t)\,dt + K_d \frac{de(t)}{dt}$ K_p : Proportional 1. Reats to present error 2. Increase speed of response according to error 3. Reduces rise time (makes sys, fast) K_i : Integral 1. Removes steady state error K_d : derivative 1. Reduces overshoot 2. Improves damping $K_d \uparrow$ noise sensitive ∴ $K_p \downarrow$ system slow $K_p \uparrow$ system oscillates	$u(t) = \text{fuzzy} = F(e, \Delta e)$ Explanation through a problem 1. desired speed : 100 rpm 2. actual motor speed : 80 rpm 3. Error = Reference − actual 100 − 80 = 20 rpm (e) 4. Assuming previous error : 25 rpm 5. Δ Error $\Delta e = e(k) - e(k-1)$ = 20 − 25 = −5 rpm e is redusing, so the motor is catching up the reference.
Disturbance rejection performance speed dip Minimum speed dip after disturbance speed dip = 19 rad/s Recovery time after apply 3 loads at 3 times ±2% tolerance T_R = 1.25 − 0.19 = 1.06 S	Disturbance rejection performance 1. Speed dip: (max) = 50 − 48 = 2 rad/s 2. Disturbance recovary 0.043 − 0.041 = 0.002 S = 2×10^{-3} S Note: fuzzy have the adaptive tunning which makes the disturbance occurs before reaching steady state with ±2% tolerance
	


EEE206Projectpid
load torque i/p2 0.2;50
load torque i/p3 0.4;90
Electromagnetic torque i/p2 100
PID(s)
PID Controller2 5;25;0.0001
num(s)
den(s)
PMSM TF2
Speed o/p2
load torque i/p4
0.2;50
load torque i/p5
0.4;60
load torque i/p1
0.6;70
Speed o/p2
File Tools View Simulation Help
160
140
120
100
80
60
40
20
0
-20
0 0.5 1 1.5 2 2.5 3 3.5 4 4.5 5
Ready
Sample based
T=5.000